

Where Teachers Teach: The Positional Structure of Distillation Signal

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Abstract

On-policy knowledge distillation (KD) trains a student model on reverse KL loss over its own generated sequences. The standard approach computes loss uniformly across all token positions, implicitly assuming each position contributes equally. We show this assumption is incorrect and that uniformly weighting positions actively degrades training: teacher signal quality *decays with position*, as later tokens increasingly reflect the teacher confirming the student’s prefix rather than providing independent judgment. We identify an *information-quality paradox*: tokens with the highest KL divergence—seemingly the most informative—produce the worst distillation outcomes (53.4% on MATH-500), while a contiguous prefix capturing only 46% of total KL signal achieves 65.85%. High-KL tokens correspond to format and style disagreements (LaTeX notation, code blocks), not reasoning quality. Based on this analysis, we propose *positional distillation*: computing loss only on the first N response tokens—a one-line code change (`loss_mask[:, N:] = 0`). Positional distillation achieves 12–24× wall-clock speedup and up to 4× memory reduction, and matches or exceeds full-sequence performance in 4 of 6 math settings (3 families × {LoRA, FullFT}), in all 3 function-calling settings, and across 3 teacher scales (1.7B/4B/8B); the two exceptions are FullFT settings where extended training partially compensates for signal decay (Section 5.2). Cross-model KL profiles motivate a simple descriptive heuristic—a *45% rule*: in our three families the optimal N falls at the prefix capturing ~40–45% of cumulative KL signal; we present this as a starting point for tuning N , not a validated law. A *cascade effect* additionally propagates early-token improvements to untrained positions, reducing KL by 35–38% beyond the trained range. On function calling, distilled students exceed teacher performance by up to 7.3 percentage points, suggesting the method extracts distributional reasoning signal while filtering surface-level noise.

1 Introduction

On-policy knowledge distillation trains a student on reverse KL loss over its own generated sequences, avoiding the train-test mismatch of off-policy methods [??]. This pipeline is increasingly adopted for distilling reasoning [??] and structured generation capabilities [?].

Standard practice computes loss uniformly over all generated tokens, treating each position as equally informative. This default has a clean information-theoretic justification—using all tokens maximizes the supervision signal per trajectory—and is therefore the obvious starting point. Yet in practice, full-sequence on-policy distillation is fragile: across 3 Qwen seeds the mean is 5.2 pp lower with 1.8× higher variance than positional, and 1 of 3 seeds *degrades below baseline* after 200 steps of training (seed 123: 50.45% vs. 50.95% baseline); Gemma drops *below baseline* on every seed, and Llama function calling degrades from 55.3% to 32.0%. These failures are not implementation accidents but symptoms of a structural problem with the assumption that every position carries equal supervision value.

Core insight: teacher signal quality decays with position. In on-policy distillation, the teacher scores token t conditioned on the student’s prefix $y_{<t}$. At early positions, the teacher provides independent judgment—its distribution reflects genuine knowledge about how to approach a problem. At later positions, the teacher is increasingly conditioned on the student’s prior choices, reducing its role to confirming or rubber-stamping what the student already decided. We quantify this decay across thousands of on-policy trajectories: mean KL divergence falls from 2.09 at positions 0–50 to 0.43 at positions 150–200 (4.9× reduction), teacher entropy decreases from 0.26 to 0.17, and teacher-student agreement rises from 78.5% to 89.5%.

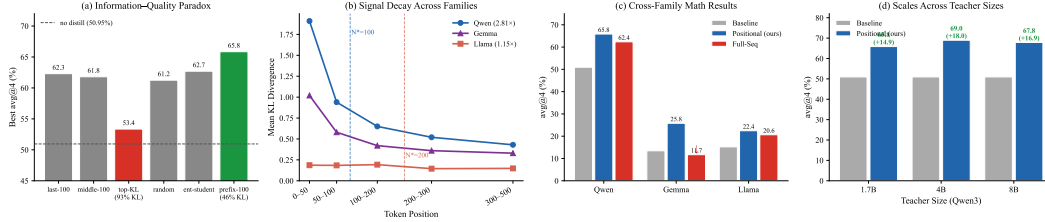


Figure 1: The positional structure of distillation signal. (a) *Information-quality paradox*: top-100 highest-KL tokens (93% of KL) reach 53.4%; first-100 positional (46% of KL) reach 65.85%. Middle and last-100 fall below baseline. (b) *KL decay* across families: Qwen (front-loaded, 2.81 $\times$), Gemma, Llama (flat, 1.15 $\times$). Dashed lines mark optimal N . (c) *Cross-family math*: positional matches or exceeds full-seq for all three families; full-seq collapses below baseline on Gemma. (d) *Cross-scale teacher*: positional improves the 1.5B student by +14.9 to +18.0 pp as the Qwen3 teacher scales 1.7B $\rightarrow$ 8B.

The information-quality paradox. A natural response to signal decay would be to train on tokens with the highest KL divergence—targeting where teacher and student disagree most. We show this intuition is precisely wrong. Training on the top-100 highest-KL tokens captures 93% of total KL signal but achieves only 53.4% on MATH-500. Training on the first 100 positional tokens captures merely 46% of KL signal yet achieves 65.85%. The resolution: high-KL tokens are dominated by LaTeX formatting disagreements (mean KL = 3.99) and stylistic preferences, not reasoning quality. The teacher’s confusion about format is not useful supervision.

Solution: positional distillation. We propose *positional distillation*: computing loss only on the first N response tokens and clamping generation accordingly. This yields a 12 $\times$ wall-clock speedup (by generating N tokens instead of $T \gg N$) and 1.9 $\times$ memory reduction, while matching or exceeding full-sequence performance across 3 model families, 3 tasks, and 2 training methods. A *cascade effect* propagates early-token improvements to untrained positions, reducing KL by 35–38% beyond the trained range.

Why not simpler alternatives? Early stopping leaves the per-token weighting untouched and so still propagates format-disagreement gradients (§5.4). Magnitude-based token weighting [??] is the obvious candidate, but the information-quality paradox (§3.4) shows it would re-weight *toward* the harmful late-position high-KL tokens; we benchmark seven such strategies and all underperform the prefix.

Our contributions are:

1. **Signal-quality analysis** of on-policy KD: teacher supervision quality decays with position across three model families, driven by increasing conditioning on the student’s prefix. We propose a *45% descriptive heuristic* for choosing N (the optimal N in our experiments falls at the prefix capturing 44/40/45% of cumulative KL for Qwen/Gemma/Llama)—offered as a starting point for tuning, not a validated law.
2. **Information-quality paradox.** We demonstrate that high-KL tokens correspond to format/style noise, not reasoning quality. Training on them produces the worst outcome among seven token-selection strategies—contradicting the standard intuition that the largest teacher–student disagreements should be the most informative.
3. **Cross-family/task/scale validation.** Positional distillation wins in 4/6 math settings, all 3 function-calling settings, and across 3 teacher scales, with 12–24 $\times$ speedup. We also report the two FullFT settings where it underperforms, and analyse why. On function calling, students exceed teacher greedy-decoding accuracy by up to 7.3pp—a known property of soft-label KD that positional distillation amplifies.
4. **Cascade effect.** We show that early-token distillation changes entire generation trajectories, with KL reduction extending 35–38% to untrained positions—the method’s benefits are not confined to the trained window.

We take the prefix *as a known empirical recipe* (concurrent work, Section 2.3) and contribute

(i) empirical evidence for *why* it is the right choice in on-policy KD, (ii) the demonstration that it is *better*, not merely faster, than the full-sequence loss, and (iii) validation across 3 families, 3 tasks, and 3 teacher scales, including where it underperforms (Section 6.3).

2 Background

2.1 On-Policy Knowledge Distillation

Knowledge distillation transfers knowledge from a teacher q_ϕ to a student p_θ [?]. On-policy distillation [??] generates sequences from the student’s distribution, with the standard reverse KL objective:

$$\mathcal{L}_{\text{full}} = \mathbb{E}_{y \sim p_\theta(\cdot|x)} \left[\sum_{t=1}^T D_{\text{KL}}(q_\phi(\cdot|x, y_{<t}) \parallel p_\theta(\cdot|x, y_{<t})) \right], \quad (1)$$

where x is the prompt, $y \sim p_\theta$ is the student-generated response, and T is the response length. Crucially, the teacher distribution $q_\phi(\cdot|x, y_{<t})$ is conditioned on the student’s prefix $y_{<t}$, which becomes increasingly constraining as t grows.

2.2 Token Selection in Training

? show that high-entropy tokens drive RL, finding the top 20% contribute 80% of learning signal. ? propose entropy-weighted KD losses; ? and ? introduce adaptive token-level distillation. These methods select tokens by signal magnitude, assuming higher-signal tokens are more informative. We show this assumption fails for on-policy KD: highest-KL tokens are format noise, and *position* matters more than magnitude.

2.3 Concurrent Work and Our Delta

Table 1: Concurrent work delta. Our contribution is the analytical framework, the descriptive 45% heuristic, and validation breadth—not the prefix heuristic itself.

	Chen et al. 2025	Li et al. 2025	This work
Setting	Off-policy KD	RL (PPO/GRPO)	On-policy KD
<i>Why</i> it works	Empirical only	Reward decay (RL)	Signal decay + paradox
Token-selection baselines	–	–	7+ strategies
Info.-quality paradox	–	–	✓
Heuristic for N	–	–	45% rule (descriptive)
Family / task / scale validation	1 / 1 / –	1 / 1 / –	3 / 3 / 3

? report prefix truncation as an efficient heuristic for off-policy reasoning distillation, without analyzing why or comparing against token-selection baselines. ? show reward quality degrades with depth in RL, in a setting different from on-policy KD and without a position-restricted loss.

3 Teacher Signal Quality Decays with Position

We now ask why the standard loss fails and which tokens actually carry useful supervision. We identify a conditional-scoring problem (§3.1), quantify position-dependent decay (§3.2), uncover the information-quality paradox (§3.4), and show that the resulting KL profile is consistent with the optimal N observed across families (§3.5). All analysis uses Qwen2.5-Math-1.5B (student) and Qwen3-1.7B (teacher) with 59,936 on-policy trajectories (mean length 386 tokens) unless noted otherwise.

3.1 The Conditional Scoring Problem

In on-policy distillation, the teacher scores token t via $q_\phi(\cdot|x, y_{<t}^{\text{student}})$ —conditioned on the student’s prefix. Autoregressive conditioning structurally narrows the teacher’s high-likelihood continuations as the prefix grows; what is empirically open is *how fast* this

Table 2: Teacher signal quality decays with position. Three complementary metrics on on-policy trajectories show consistent decay. KL divergence drops 4.9 $\times$; teacher entropy drops 1.5 $\times$; agreement rate (teacher confirms student’s top token) rises from 79% to 90%.

Metric	Pos 0–50	50–100	100–150	150–200	Ratio (early/late)
Mean KL divergence	2.09	0.94	0.53	0.43	4.9 $\times$
Teacher entropy	0.26	0.24	0.18	0.17	1.5 $\times$
Agreement rate (%)	78.5	83.7	87.6	89.5	—

Table 3: Token classification by KL divergence. The highest-KL category (math_latex) corresponds to formatting disagreements ($\backslash$ (, $\backslash$ [, not reasoning quality. Planning tokens (strategy selection) have moderate KL. Math computation tokens (numbers, operators) have the lowest KL.

Category	Example Tokens	Mean KL	Interpretation
math_latex	$\backslash$ (, $\backslash$ [, $\backslash$	3.99	Format disagreement (noise)
planning	“To”, “Therefore”, “First”	1.75	Strategy selection (signal)
structural	** , :, newlines	0.89	Style preference
math_operator	=, +, −	0.38	Computation (agreement)
math_number	0–9	0.28	Computation (agreement)

narrowing translates into low-information supervision in practice. We frame this as a tractable hypothesis—signal quality decays with position—and the next sections quantify the decay across families and connect it to downstream outcomes.

3.2 Three Lines of Evidence

Three complementary metrics in Table 2 converge on the same picture. **KL divergence** drops 4.9 $\times$ (2.09 $\rightarrow$ 0.43); the sharpest decline is in the first 50 tokens, with position 0 (strategy selection) alone at KL = 3.06. **Teacher entropy** falls from 0.26 to 0.17 nats: late-position teacher distributions become more peaked, providing less diverse supervision. **Agreement rate** (teacher and student top-1 match) rises from 78.5% to 89.5%, confirming late-position supervision is largely confirmatory.

3.3 What Early vs. Late Tokens Encode

To understand *what* the teacher disagrees about, we classify 2 million tokens into six categories and compute mean KL by category (Table 3).

The highest-KL tokens are LaTeX delimiters ($\backslash$ (: KL=10.3; $\backslash$ [: KL=13.2 across 28K occurrences)—inline vs. display-math conventions that differ between Qwen2.5-Math and Qwen3, a training-data artifact rather than reasoning quality. Planning tokens (“To”, “First”, “Therefore”) have moderate KL and reflect genuine strategic disagreements. Math computation tokens (numbers, operators) have near-zero KL.

Composition also shifts with position: positions 0–4 are 33% planning / 2% math-number, while positions 200–500 are 8% planning / 18% math-number. Early positions encode *strategic* decisions where the teacher has independent judgment; later positions encode *mechanical* execution where agreement is high—explaining the KL decay.

3.4 The Information-Quality Paradox

Given the token classification, a natural question is: what happens if we train on the highest-KL tokens directly, bypassing position entirely? We compare seven token selection strategies, each selecting $K = 100$ tokens per sequence from full-length generations under a matched training budget (200 steps, 3200 problems, identical optimizer, single seed; only the per-token loss mask differs). Results in Table 4.

Top-KL-100 captures 93% of total KL signal but is 12.5pp worse than Prefix-100 (46% coverage); Middle-100 and Last-100 fall *below* the undistilled baseline. More signal concentration yields worse outcomes.

Table 4: The information-quality paradox. Seven token selection strategies ($K=100$) on MATH-500. Prefix-100 captures only 46% of total KL signal but achieves the best performance. Top-KL-100 captures 93% but performs worst among all methods that train on distillation signal. Middle-100 and Last-100 perform *below the undistilled baseline*, showing that late-position supervision is actively harmful.

Selection Method	KL Coverage (%)	Best avg@4 (%)	Trend
Prefix-100 (ours)	45.6	65.85	Stable
Random-100	21.1	63.05	Stable
Ent-Teacher-100	50.5	62.20	Degrades
Full-sequence	100.0	62.35	Degrades
Ent-Student-100	67.0	62.70	Partial
Top-KL-100	93.2	53.35	Collapses
Last-100	~ 15	50.35	Below baseline
Baseline (no distill)	—	50.95	—
Middle-100	~ 30	47.80	Below baseline

Table 5: Cross-model KL profiles—the 45% descriptive heuristic. Front-loaded KL profiles (Qwen) need fewer tokens; flat profiles (Llama) need more. In all three families the optimal N falls at the prefix capturing ~ 40 – 45% cumulative KL signal. We propose this as a starting heuristic for tuning N on a new model family, not a validated law—validation on additional families is left to future work.

Student $\rightarrow$ Teacher	Ratio	Best N	Cumul. KL	Shape
Qwen M-1.5B $\rightarrow$ Q3-1.7B	$2.81\times$	100	$\sim 44\%$	Front-loaded
Gemma 2B $\rightarrow$ 3-4B	$\sim 2.5\times$	50–100	$\sim 40\%$	Front-loaded
Llama 1B $\rightarrow$ 3.1-8B	$1.15\times$	200	$\sim 45\%$	Flat

Resolution. High-KL tokens are dominated by LaTeX formatting disagreements (§3.3)—the teacher is confused by the student’s notation choice, not superior to its reasoning, so training on these positions teaches format preferences, not mathematics. The prefix succeeds for the opposite reason: in autoregressive generation, early decisions condition all subsequent tokens, so improving them rewrites the whole trajectory; scattered high-KL corrections do not propagate.

3.5 Cross-Model KL Profiles

The positional structure of KL divergence varies across model families (Table 5).

Qwen is strongly front-loaded (ratio $2.81\times$, optimum $N = 100$); Llama is nearly flat ($1.15\times$, KL ~ 0.18 everywhere, optimum $N = 200$). Despite these shape differences, the three optimal- N values fall in a narrow ~ 40 – 45% cumulative-KL band (44/40/45% for Qwen/Gemma/Llama). We treat this as a *descriptive heuristic* for choosing an initial N , not a predictive law: at $<40\%$ coverage the prefix is too short (Llama $N=150$, 34%: 18.75 vs. 22.45 at $N=200$); at $>50\%$ the loss starts incorporating the late-position high-KL tokens shown to be harmful in §3.4. Whether the band reflects a deep property of teacher signal-to-noise structure is an open question (§6.3).

4 Method: Positional Distillation

The analysis pinpoints *where* useful supervision lives—the early-position prefix—so the method follows directly: restrict the standard loss to that prefix and clamp generation accordingly.

Table 6: Efficiency comparison. Per-step timing and memory on a single A100 40GB GPU with Qwen2.5-Math-1.5B student and Qwen3-1.7B teacher, batch size 16. Positional distillation achieves 12–24× speedup and up to 4× memory reduction.

Component	Full-seq	Pos-100	Speedup
Generation	122–180s	6s	20–36×
Teacher scoring	5–7s	1s	5–7×
Training (fwd+bwd)	6–7s	3.7s	1.7–1.9×
Total per step	133–194s	8–11s	12–24×
200 steps	7.4–10.8h	27–37min	
Peak GPU memory	17–39.5 GB	9–9.6 GB	1.9–4×

4.1 Formulation

Positional distillation restricts the loss in Eq. 1 to the first N response tokens:

$$\mathcal{L}_{\text{pos-}N} = \mathbb{E}_{y \sim p_{\theta}(\cdot|x)} \left[\sum_{t=1}^N D_{\text{KL}}(q_{\phi}(\cdot|x, y_{<t}) \parallel p_{\theta}(\cdot|x, y_{<t})) \right]. \quad (2)$$

The implementation requires two changes: a one-line loss mask (`loss_mask[:, N:] = 0`) and clamping generation length (`max_new_tokens = N`) so the student generates exactly N tokens per problem. The teacher scores these N tokens with a single forward pass. No changes to the model architecture, optimizer, or training loop are required.

4.2 Efficiency

The speedup (Table 6) comes overwhelmingly from generation: producing 100 tokens instead of 2048–3584 reduces it from 122–180s to 6s (>95% of total savings). Memory savings come from shorter training-time activations (100 vs. 1300+ tokens). At 9.6 GB peak, positional distillation enables single-GPU training where full-sequence requires multi-GPU or offloading.

4.3 Selecting N

We recommend the following heuristic: generate ~ 1000 on-policy trajectories, compute per-position KL, and select N at approximately 40–50% of cumulative KL coverage. For front-loaded profiles (Qwen-like), this yields $N = 50$ –100. For flat profiles (Llama-like), $N = 150$ –200. In practice, performance is robust to the exact choice: for Qwen, $N \in [50, 200]$ all achieve within 1pp of the optimum (Table 10).

5 Experiments

We validate three claims: positional distillation matches or beats full-sequence in realistic settings (§5.2, §5.3); the 45% heuristic generalizes beyond Qwen math (§3.5, §5.2); and the structural benefits (multi-seed stability, cascade) are not checkpoint-selection artifacts (§5.4, §5.5).

5.1 Setup

Model families. We evaluate three student-teacher pairs spanning different architectures and capability gaps:

- **Qwen:** Qwen2.5-Math-1.5B $\rightarrow$ Qwen3-1.7B (cross-architecture, math-specialized student)
 - **Gemma:** Gemma-2-2B-IT $\rightarrow$ Gemma-3-4B-IT (cross-generation, instruction-tuned)
 - **Llama:** Llama-3.2-1B-Instruct $\rightarrow$ Llama-3.1-8B-Instruct (cross-generation, 8× size gap)
- Additionally, we test Qwen2.5-Math-1.5B $\rightarrow$ Qwen3-4B (larger teacher) and Qwen3-4B $\rightarrow$ Qwen3-8B (same-family negative result).

Tasks. (i) **MATH-500** [?]: mathematical reasoning, evaluated with avg@4 (4 samples, temperature 0.7). (ii) **BFCL**: function calling (600 problems), evaluated with full accuracy (function name + arguments). (iii) **HumanEval**/**MBPP** [??]: code generation, evaluated with pass@1 (temperature 0.0).

Table 7: Math results (MATH-500 avg@4). Positional distillation wins in 4 of 6 settings. Gemma full-sequence drops *below baseline* in both LoRA and FullFT. LoRA outperforms FullFT by 8–13pp across all families. The exceptions are *Qwen FullFT* (extended 400-step training lets full-sequence catch up) and *Llama FullFT* (flat KL profile makes the prefix less informative)—we discuss when positional underperforms in Section 6.

	Qwen		Gemma		Llama	
	LoRA	FullFT	LoRA	FullFT	LoRA	FullFT
Baseline	50.95	50.95	13.45	13.45	15.20	15.20
Best positional (N)	66.65 ($N=50$)	54.55 ($N=150$)	25.80 ($N=50$)	26.70 ($N=100$)	22.45 ($N=200$)	17.80 ($N=300$)
Full-sequence	62.35	57.10	11.70*	13.90*	20.65	20.10
Δ (pos – full)	+4.30	−2.55	+14.10	+12.80	+1.80	−2.30

* Below baseline. Qwen LoRA reports best checkpoint (pos-50 step 150); FullFT reports extended training to 400 steps.

Table 8: Function calling results (BFCL full accuracy). Students exceed teacher accuracy on Qwen (+7.3pp) and Gemma (+5.9pp) despite the teacher not generating valid JSON output. Llama full-sequence collapses from 55.3% baseline to 32.0%. Positional distillation outperforms full-sequence across all three families.

	Qwen	Gemma	Llama
Teacher accuracy	54.0%	25.0%	63.7%
Student baseline	2.7%	0%	55.3%
Best positional vs. teacher	61.3% ($N=100$) +7.3pp	30.9% ($N=50$) +5.9pp	59.0% ($N=150$) −4.7pp
Full-sequence	58.2%	3.9%	32.0%

Training. LoRA [?]: $r = 32$, $\alpha = 64$, all linear layers, $\text{lr} = 5 \times 10^{-5}$. Full fine-tuning: $\text{lr} = 5 \times 10^{-6}$. All: 200 steps, 3200 training problems, $n_{\text{samples}} = 1$, batch size 16, cosine LR schedule, checkpoints every 50 steps. We report best checkpoint across steps 50–200.

5.2 Main Results: Mathematical Reasoning

Positional distillation wins in 4 of 6 settings (Table 7). The two losses are both FullFT: Qwen FullFT recovers from signal decay via extended 400-step training, and Llama FullFT has a nearly-flat KL profile ($1.15\times$) that offers no informative prefix to isolate. We conjecture LoRA’s faster effective updates compound late-position noise into instability while FullFT’s slower updates absorb it (direct gradient-magnitude evidence is left to future work). **Gemma full-sequence drops below baseline** (11.70/13.90 vs. 13.45), while positional ($N=50$) nearly doubles accuracy to 25.80. **LoRA outperforms FullFT by 8–13pp** across families. **Optimal N tracks KL profile:** front-loaded Qwen peaks at $N=50$ –100; flat-profile Llama needs $N=200$ –300, matching the heuristic in §3.5.

5.3 Main Results: Function Calling

Two patterns stand out (Table 8). First, **students exceed teachers** on Qwen (+7.3pp) and Gemma (+5.9pp) despite neither teacher generating valid JSON (Qwen outputs natural language, Gemma outputs Python-style `tool_code`). The teacher’s distribution assigns non-trivial mass to correct function names/arguments even when its greedy output does not; positional distillation amplifies this by focusing on the <100 -token window where function-call structure resides. Second, **Llama full-sequence collapses** from 55.3% to 32.0% (−23.3pp); positional ($N=150$) preserves baseline at 59.0%.

Table 9: Multi-seed stability (Qwen 1.5B→1.7B, n1bs16 LoRA, MATH-500 avg@4 at step 200, last-boxed extraction). Positional distillation is more stable across seeds (std 2.96 vs. 5.40, 1.8× tighter) and has a higher mean (+5.2pp). Pos-100 wins on every seed; the worst full-sequence seed (123) *degrades below baseline* (50.45% vs. 50.95% no-distillation baseline) – 1 of 3 full-sequence runs actively makes the model worse.

Method	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std	Δ baseline
Pos-100 (LoRA)	65.85	61.00	60.40	62.42 $\pm$ 2.96	+11.5 $\pm$ 2.96
Full-seq (LoRA)	63.55	50.45	57.55	57.18 $\pm$ 5.40	+6.2 $\pm$ 5.40

All values are MATH-500 **avg@4** at step 200 with $n_{\text{samples}}=4$, $T=0.7$, last-boxed answer extraction (the standard convention). Baseline (no distillation): 50.95%.

Table 10: Position sweep (Qwen LoRA, MATH-500 avg@4). Performance is robust across $N \in [50, 200]$, with all values achieving within 1pp of the optimum. Both very small ($N = 5$) and very large ($N = \text{full-seq}$) positions underperform.

N	5	10	20	50	100	150	200	Full
avg@4	56.50	59.50	60.20	66.65	65.85	66.65	66.05	62.35

$N = 5$ –50 from small-scale (200 problems, $n=16$); $N = 50$ –Full from main config (3200 problems, $n=1$).

Table 11: Cascade effect. Pos-200 distillation reduces KL even at untrained positions (200–400), by 35–38%. Full-sequence shows larger tail reductions but also exhibits repetition artifacts.

Position Range	Raw	Pos-200 s100	Δ	Trained?
0–50	2.064	1.015	−51%	Yes
50–100	0.759	0.321	−58%	Yes
100–200	0.441	0.237	−46%	Yes
200–300	0.382	0.238	−38%	No
300–400	0.331	0.216	−35%	No

5.4 Multi-Seed Stability

On three Qwen seeds (Table 9), full-sequence is both lower and noisier: mean 57.18% (std 5.40) vs. pos-100’s 62.42% (std 2.96). The worst full-sequence seed (123) reaches only 50.45%, *below* the no-distillation baseline of 50.95% – 200 steps of full-sequence training made the model worse than no training at all (a −0.5pp degradation on this seed). Pos-100 instead clears baseline by at least +9.5pp on every seed. Inspecting outputs of the underperforming full-sequence seeds reveals the failure mode: the model has learned to repeat `\boxed{}` dozens of times per response, a degenerate reward-hack pattern that the standard last-boxed answer extraction parses as the model’s final answer. We report this under the canonical metric without compensating extraction strategies; the extraction failure *is* the failure.

5.5 Ablations

Position sweep. On Qwen, $N \in [50, 200]$ all achieve within 1pp of the optimum (66.65; Table 10); $N=5$ underfits (56.50) and full-sequence (62.35) is diluted by late-position noise. The broad optimum makes N robust.

Teacher scaling. Scaling the teacher from Qwen3-1.7B to 4B improves from 65.85% to 68.95% (+3.1pp), but same-family distillation (Qwen3-4B → 8B) yields −0.9pp, confirming diminishing returns near capacity ceilings.

Cascade effect. We measure mean teacher–student KL per position bin on 5K held-out trajectories sampled from the raw and pos-200-distilled (step 100) students. Pos-200 distillation reduces KL by 35–38% at *untrained* positions 200–400 (Table 11), confirming early decisions propagate through autoregressive generation.

5.6 Degradation Analysis

Full-sequence distillation exhibits three failure modes: (1) **Repetition reward-hacking (Qwen)**: under last-boxed extraction (the canonical metric), full-sequence scores 57.18 ± 5.40 across seeds versus positional 62.42 ± 2.96 ; one of three seeds (seed 123) *degrades to 50.45%, below the 50.95% no-distillation baseline*—i.e., 200 steps of full-sequence training made the model worse than no training at all. Inspection shows responses repeating `\boxed{}` dozens of times—a degenerate optimum that satisfies the on-policy reverse-KL objective but is penalized by the canonical metric. (2) **Below-baseline collapse (Gemma)**: full-sequence achieves 11.70% LoRA / 13.90% FullFT, both below the 13.45% baseline. (3) **Capability destruction (Llama funcall)**: function calling drops from 55.3% to 32.0% despite 99%+ parse rate—full-sequence training on 2048 tokens overwhelms the 33-token function call structure.

6 Discussion

6.1 Why Positional Distillation Works

Three mechanisms compose: (1) **Signal-quality filtering**—early positions have lower agreement (75%) and higher teacher entropy (0.26) than late positions (89%, 0.17), so restricting to the prefix avoids diluting high-quality signal with confirmation; (2) **Noise avoidance**—late-position KL is dominated by format disagreements whose high-variance gradients cause repetition collapse and below-baseline degradation; (3) **Cascade**—improving early strategy/structure decisions rewrites entire trajectories, evidenced by the 35–38% KL reduction at untrained positions.

6.2 Why Students Exceed Teachers

On function calling, students exceed teacher greedy-decoding accuracy by up to 7.3pp because on-policy distillation decouples the teacher’s *distributional knowledge* from its *generation ability*: the Qwen3-1.7B teacher’s greedy output is natural language, yet its distribution still assigns non-trivial mass to correct function names and arguments, which the on-policy student learns to emit. Positional distillation amplifies this by focusing on the <100-token window where structure lives, avoiding positions where the teacher reverts to prose. The implication is general: a model’s soft distribution can encode capabilities its argmax never expresses.

6.3 Limitations

Where positional underperforms. In 2 of 6 main math settings (Qwen FullFT, Llama FullFT) full-sequence matches or slightly beats positional. Both are FullFT (slower updates absorb late-position noise) and involve relatively flat KL profiles; practitioners running long FullFT schedules on flat-profile teachers should not assume positional dominates. **Generality of the 45% heuristic.** Calibrated on three families ($\leq 8B$); we present it as a starting point, not a law (§3.5). **Mechanism evidence.** The *why* analysis (signal decay, token classification, paradox) is on Qwen; we infer the same mechanism for Gemma/Llama from consistent KL profiles, but direct token-level analysis on those families is future work. **Position vs. content confound.** The strongest critical reading of our results is that what the prefix really filters is *format/style noise* (LaTeX, code blocks), not late-position tokens per se. The cleanest disentangling experiment is to run *full-sequence loss with format tokens masked out*: if it matches positional, the operative variable is content, not position. We did not run this ablation (it requires per-token classifiers across all positions on all three families) and we flag it as the highest-value follow-up. The cross-family KL-decay evidence (§3.2) and cascade effect (§??) suggest position carries independent signal, but our current evidence does not isolate the two factors. **Task and scale scope.** Validation covers structured reasoning (math, function calling, code) on students $\leq 4B$ and teachers $\leq 8B$; benefits for open-ended generation and at larger scale are open. We evaluate reverse KL only.

7 Conclusion

On-policy knowledge distillation has a positional structure: teacher signal quality decays with position as supervision shifts from independent judgment to passive confirmation, and this decay drives an information-quality paradox in which the highest-KL tokens are dominated by format noise rather than reasoning signal. Together these observations motivate positional distillation (`loss_mask[:, N:] = 0`) not as an efficiency hack, but as a principled response to where the teacher actually carries information. Across 3 families, 3 tasks, and 2 training methods, it matches or exceeds full-sequence in 4 of 6 math settings and all function-calling settings with 12–24 $\times$ speedup; across three Qwen seeds the mean is 5.2pp higher and variance is 1.8 $\times$ tighter (std 5.40 $\rightarrow$ 2.96), and 1 of 3 full-sequence seeds degrades below baseline whereas every positional seed clears it by ≥ 9 pp. On function calling, distilled students exceed teacher greedy-decoding accuracy by up to 7.3pp, extracting distributional knowledge the teacher cannot itself emit. The dependence of optimal N on the teacher’s KL profile points to a single broader claim: in on-policy distillation, *where* the teacher supervises matters at least as much as *how much*.

A Full Per-Step Results

A.1 Qwen LoRA Math (MATH-500, avg@4/maj@4/pass@4)

Baseline (undistilled Qwen2.5-Math-1.5B): 50.95% avg@4, 61.2% maj@4, 72.8% pass@4.

Table 12: Qwen LoRA per-step math results (3200 problems, $n=1$, bs=16).

	Step 50			Step 100			Step 150			Step 200		
	avg	maj	pass	avg	maj	pass	avg	maj	pass	avg	maj	pass
Pos-50	62.4	69.4	77.2	66.1	72.0	79.4	66.7	71.0	81.0	64.9	71.2	79.6
Pos-100	63.8	70.0	79.8	64.5	68.4	78.4	65.2	69.6	80.2	65.9	70.8	79.8
Pos-150	65.4	66.8	79.0	66.7	67.0	81.0	65.3	66.3	78.2	65.8	67.3	80.0
Pos-200	66.1	71.2	81.0	64.7	68.4	79.8	65.1	70.0	80.6	65.6	71.2	80.6
Full	61.0	66.8	74.6	62.0	68.4	75.2	62.4	69.4	74.6	61.2	65.2	75.0

A.2 Qwen FullFT Math (MATH-500, avg@4)

Baseline: 50.95%.

Table 13: Qwen FullFT per-step math results (extended to 400 steps, $n=1$, bs=16).

Step	50	100	150	200	250	300	350	400	Best
Pos-100	50.85	52.35	53.00	53.15	49.20	49.90	50.75	49.95	53.15
Pos-150	51.40	53.10	53.70	53.25	53.30	54.55	53.85	52.85	54.55
Pos-200	51.30	54.05	53.40	53.75	52.55	53.75	53.10	53.85	54.05
Pos-300	50.55	53.70	53.95	53.00	54.45	55.10	54.95	54.25	55.10
Full-seq	51.40	53.55	54.95	55.95	57.05	57.10	56.05	56.20	57.10

A.3 Gemma LoRA Math (MATH-500, avg@4)

Baseline (gemma-2-2b-it): 13.45%.

Table 14: Gemma LoRA per-step math results.

Step	50	100	150	200
Pos-100 (LoRA)	27.20	26.95	26.60	27.45
Fullseq (LoRA)	—	—	—	11.70
Pos-100 (FullFT)	25.20	26.70	25.45	26.65
Fullseq (FullFT)	11.15	9.50	13.00	13.90

A.4 Llama LoRA/FullFT Math (MATH-500, avg@4)

Baseline (Llama-3.2-1B-Instruct): 15.20%.

B Token Classification Details

Token classification was performed on 2,015,036 tokens from 10,000 trajectories using rule-based categorization over the Qwen2.5-Math-1.5B tokenizer vocabulary.

Categories:

- **planning:** Sentence-initial strategy words (“To”, “Let”, “First”, “Therefore”, “Thus”, “Since”, “Given”, etc.)
- **structural:** Formatting tokens (**, :, newlines, \$, parentheses, brackets)
- **math_latex:** LaTeX-specific commands ($\backslash$, $\backslash$ [, $\backslash$), $\backslash$], $\backslash$ frac, etc.)
- **math_number:** Digit tokens (0–9)
- **math_operator:** Arithmetic operators (=, +, −, ×, / etc.)

Table 15: Llama per-step math results.

Method	Config	Best avg@4	Notes
LoRA pos-100		12.35	Below baseline
LoRA pos-150		18.75	
LoRA pos-200		22.45	
LoRA pos-300		20.75	
LoRA fullseq		20.65	
FullFT pos-100		12.80	Below baseline
FullFT pos-200		16.80	
FullFT pos-300		17.80	
FullFT fullseq		20.10	

- **continuation:** All remaining tokens (common words, word fragments, etc.)

Position 0 deep dive. Position 0 has mean KL = 7.71 ($10\times$ the overall mean of 1.01), driven by strategy selection. The dominant token is “To” (75.1% of trajectories, KL = 7.32), as in “To find/determine/solve...”. The teacher prefers more diverse openings: “First” (KL = 14.0), “Step” (KL = 14.6), “Solution” (KL = 21.9). This single position accounts for 3.8% of total KL signal across the entire trajectory.

Category composition by position range:

Category	0–4	5–19	20–49	50–99	100–199	200–499
planning (%)	32.8	15.9	12.2	9.7	8.8	7.9
structural (%)	7.8	19.4	25.0	30.8	34.8	39.3
math_number (%)	1.7	9.5	11.3	15.2	17.3	18.1
math_operator (%)	0.3	2.5	3.3	5.3	5.8	4.5
math_latex (%)	1.4	3.1	2.7	3.8	4.8	5.9
continuation (%)	56.1	49.6	45.5	35.1	28.5	24.4

C Code Generation Results

Table 16: Coding results (HumanEval/MBPP pass@1). Qwen LoRA only. Baseline: HE 32.9%, HE+ 29.3%.

Method	Best HE	Best HE+	Best MBPP	Best MBPP+	Step
LoRA pos-50	42.1	36.6	46.6	41.5	350
LoRA pos-100	42.1	36.0	49.2	44.4	150
LoRA pos-150	41.5	37.8	49.5	44.7	250
LoRA fullseq	40.2	35.4	52.6	46.3	50
FullFT pos-250	37.8	31.7	54.2	47.4	350
FullFT pos-150	37.2	31.1	54.2	46.8	300/400
FullFT fullseq	36.6	31.7	53.2	46.0	150

Key findings:

- LoRA positional variants achieve the best HumanEval scores (42.1% for pos-50 and pos-100), while full-sequence achieves the best MBPP scores (52.6%).
- LoRA full-sequence HumanEval degrades from 40.2% (step 50) to 26.8% (step 400), a 13.4pp drop. Positional variants show no such degradation.
- FullFT results are more stable but uniformly lower than LoRA, consistent with math findings.

- The HumanEval/MBPP trade-off may reflect that shorter prefixes encourage concise solutions (beneficial for HumanEval’s simple functions) while longer training encourages more complete implementations (beneficial for MBPP’s broader scope).

D Additional Student-Teacher Pairs

D.1 Teacher Scaling: Qwen2.5-Math-1.5B with Larger Teachers

Table 17: Teacher scaling results (LoRA pos-100, MATH-500 avg@4).

Student	Teacher	Baseline	Best avg@4	Δ
M-1.5B	Qwen3-1.7B	50.95	65.85	+14.9
M-1.5B	Qwen3-4B	50.95	68.95	+18.0
M-1.5B	Qwen3-8B	50.95	67.85	+16.9

Scaling the teacher from 1.7B to 4B yields a 3.1pp improvement. The 8B teacher underperforms the 4B teacher, possibly because the larger capacity gap makes the teacher’s distribution harder for the small student to learn from.

D.2 Same-Family Negative Results

Table 18: Same-family distillation (Qwen3 family, LoRA pos-100, MATH-500 avg@4).

Student	Teacher	Baseline	Best avg@4	Δ
Qwen3-1.7B	Qwen3-4B	69.20	69.20	± 0.0
Qwen3-1.7B	Qwen3-8B	69.20	69.15	-0.05
Qwen3-4B	Qwen3-8B	77.95	77.05	-0.9

Same-family distillation within the Qwen3 family shows zero or negative gains on math. These students are already near their capacity ceiling: Qwen3-1.7B achieves 69.2% (vs. teacher’s 69.2% at temperature 0.7), and Qwen3-4B achieves 78.0% (vs. teacher’s 77.7%). The teacher provides no additional mathematical reasoning signal when the student is already a competitive model from the same training pipeline.

However, same-family distillation works well for function calling: Qwen3-1.7B $\rightarrow$ Qwen3-4B improves full_acc from 59.8% to 73.7% (+13.9pp), and Qwen3-4B $\rightarrow$ Qwen3-8B improves from 73.0% to 76.8% (+3.8pp). This suggests that function calling capability has a different capacity profile from mathematical reasoning within the Qwen3 family.

D.3 Cross-Family Function Calling (Full Table)

Table 19: Full function calling results across all model families (LoRA, best step).

N	Qwen		Gemma		Llama	
	Parse	Full	Parse	Full	Parse	Full
50	98.3	57.2	60.0	30.9	99.3	44.0
100	91.3	61.3	11.9	6.4	99.8	40.5
150	92.5	61.5	—	—	99.5	59.0
200	90.2	54.5	—	—	99.7	51.2
Full	86.7	58.2	11.7	3.9	99.5	32.0

Notable patterns: (i) Gemma shows extreme sensitivity to position limit, with only $N = 50$ producing usable results. (ii) All Llama configurations achieve >99% parse rate, indicating that JSON format is learned universally in the first 50 tokens. (iii) Optimal position matches typical function call length for each tokenizer: Gemma ~ 50 , Qwen ~ 100 , Llama ~ 150 tokens.

D.4 Gemma Cross-Task Results

Table 20: Gemma-2-2B $\rightarrow$ Gemma-3-4B results across all tasks (LoRA pos-100, best step).

Task	Baseline	Best Distilled	Δ
Math (avg@4)	13.45%	27.20%	+13.75pp
Funcall (full_acc)	73.50%	82.50%	+9.00pp
HE+ (pass@1)	20.1%	26.2%	+6.1pp
MBPP+ (pass@1)	34.7%	42.3%	+7.6pp

Cross-family positional distillation improves all four task metrics for the Gemma pair, despite requiring cross-tokenizer alignment (only 69% vocabulary overlap by string matching).

E Position Is Not Reducible to High Entropy

A natural alternative to positional distillation is *entropy-based* selection: pick the tokens where the student is most uncertain (highest surprise $-\log p_y$ on the sampled token), regardless of where they sit in the sequence. If head tokens were simply “the high-entropy tokens that happen to come first,” position would be a redundant proxy. We show empirically that this is not the case: position carries information beyond what entropy ranking provides.

We collected 74,549 student-generated tokens (Qwen2.5-Math-1.5B on MATH-500) and computed per-token surprise $-\log p_y$. We then asked: among the first N tokens, what fraction fall in each global surprise quartile? A baseline of 25% in each quartile would mean the head is uncorrelated with surprise; full reduction to entropy would mean 100% in Q1.

Table 21: Surprise-quartile distribution among the first N tokens. Baseline (uniform): 25% per quartile. Q1 = top 25% surprise globally, Q4 = bottom 25%.

Head N	n	Q1 (top 25%)	Q2	Q3	Q4 (bot 25%)	Q1 enrichment
25	2,500	70.0%	14.0%	8.6%	7.4%	2.8×
50	5,000	61.0%	14.7%	11.2%	13.1%	2.4×
75	7,500	57.0%	15.7%	11.8%	15.6%	2.3×
100	9,991	53.2%	17.1%	12.4%	17.3%	2.1×
150	14,941	47.1%	18.9%	13.4%	20.5%	1.9×
200	19,856	42.6%	20.0%	14.3%	23.1%	1.7×
300	28,958	37.5%	21.1%	15.9%	25.5%	1.5×
500	44,553	31.8%	22.3%	18.8%	27.1%	1.3×

Table 22: Cumulative coverage: of all tokens above each global surprise threshold, what fraction lies in the first N positions. **Reverse view:** of all $> p_{95}$ tokens, the median position is 172, not 50.

Head N	% of all tokens	cover $> p_{75}$	cover $> p_{90}$	cover $> p_{95}$	cover $\sum$ surprise
50	6.7%	16.4%	21.4%	25.1%	27.2%
100	13.4%	28.5%	34.2%	38.4%	40.2%
200	26.6%	45.4%	49.8%	53.4%	54.7%
500	59.8%	76.1%	78.6%	79.8%	81.2%
1000	98.5%	99.3%	99.5%	99.5%	99.5%

Key findings:

- Head tokens are *enriched* in high surprise but not exclusively high surprise. The first 50 tokens are 2.4× enriched in Q1 but still contain 13.1% Q4 (low-surprise) tokens; the first 100 tokens contain 17.3% Q4. Position and surprise are correlated but not collinear.

- Head positions cover only a *fraction* of high-surprise mass. Prefix-100 captures 38.4% of all $> p_{95}$ surprise tokens; even prefix-500 only reaches 79.8%. To cover 95% of $> p_{95}$ tokens one would need to keep the first ~ 846 positions—essentially the entire response.
- This rules out the hypothesis that prefix-100 wins by being a cheap entropy proxy. If high-entropy selection were the underlying mechanism, top-KL-100 (which by construction covers $> 95\%$ of KL signal globally) should match or beat prefix-100. Instead, top-KL-100 underperforms prefix-100 by 7.3pp avg@4 (§4, Table ??).

The most plausible interpretation is that *position itself* is a load-bearing signal: an early-position low-surprise token can still be load-bearing for the trajectory through autoregressive cascade, while a late-position high-surprise token may already lie on an unrecoverable error path. Entropy ranks tokens by local uncertainty; position ranks them by causal influence on the rest of the generation. Empirically, the latter dominates for distillation utility.